

Status

HelixCode — Feature Status

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Per-feature status ledger (§11.4.153). Every user-facing capability of the HelixCode stack, its test-type coverage (§11.4.169), validation verdict, and the captured-evidence path that backs it. Validation vocabulary: PASS / SKIP / OPERATOR-BLOCKED / PENDING (§11.4.45). Evidence lives under the git-ignored qa-results/ tree (regenerated by re-running the suite, §11.4.77); curated copies for this release are under docs/qa/<run-id>/ (§11.4.83).

Components

- **Caddy edge** — TLS termination + HTTP→HTTPS redirect + reverse proxy (host 52443/52080).
- **code-server** — browser VS Code, auth-gated, projects bind-mounted RW.
- **helix-auth** (*new, in progress*) — host-native, non-root forward-auth gate behind Caddy; issues a signed SSH-key challenge and verifies the pasted signature against the real account's `authorized_keys` (`ssh-keygen -Y verify`), issues the session cookie, fails closed. No password stored.
- **Operator scripts** — `install` / `setup` / `start` / `stop` / `restart` / `status` / `logs` / `doctor` / `set-password` / `tune-host` / `(un)install-service`.
- **Boot-survival service** — rootless systemd user unit (`linger`, `WantedBy=default.target`).

Feature ledger

Component	Feature	Test types (§11.4.169)	Validation	Evidence
Caddy edge	TLS handshake / secure edge	integration, security,	PASS	tests/{integration

Component	Feature	Test types (§11.4.169)	Validation	Evidence
		tls_letsencrypt, challenges		
Caddy edge	HTTP→HTTPS redirect enforced	security	PASS	tests/secur
Caddy edge	Let's Encrypt issuance + auto-renew + rotation	tls_letsencrypt (Pebble local CA)	PASS	tests/tls_l (issuer=Pe
Caddy edge	Real public LE (public domain / DNS-01)	—	OPERATOR-BLOCKED	docs/guides token)
code-server	Login auth (correct→302+cookie, wrong→200)	e2e, security, challenges, helixqa	PASS	tests/{e2e,
code-server	Authenticated editor served	e2e, helixqa	PASS	tests/{e2e,
code-server	Projects read-write	integration, challenges, helixqa	PASS	tests/{inte round-trip)
code-server	In-browser editing (type/save via UI)	helixqa (HC-QA-UI-001)	OPERATOR-ATTENDED	tracked HC adapter)
code-server	file-watcher fix (inotify/watcherExclude)	unit, integration, challenges + guard	PASS	tests/test_ (CH4)
Operator scripts	set-password atomic .env rewrite	unit, concurrency	PASS	tests/{unit file)
Operator scripts	install → boot-survival service	(live install verified)	PASS	install log - Linger=yes
Stack	resilience (flood / kill-recover / fd pressure)	load, stress_chaos	PASS	tests/{load
Stack	performance / memory bounds	benchmark, memory	PASS	tests/{benc RSS<60%)
Stack	no secret leak / rootless	security	PASS	tests/secur podman)

Real-account authentication (new — in progress, §11.4.6)

These capabilities are **under construction** per the auth-pivot spec ([docs/superpowers/specs/2026-07-01-auth-pivot-ssh-key.md](https://docs.superpowers/specs/2026-07-01-auth-pivot-ssh-key.md), which supersedes the PAM-login part of the original design; user guide docs/guides/AUTH.md). They are **not yet validated** — tests and evidence are being built by other work streams, so

Implementation is **In progress**, Validation is **PENDING**, and Video-confirmation is **PENDING**. Do **not** read these rows as passing.

Component	Feature	Test types (§11.4.169)	Validation	Evidence
helix-auth	Real-account SSH-key login (challenge-response, verified vs authorized_keys, no password stored)	unit, integration, e2e, security, challenges, helixqa	PENDING	Impl: In progress · Video: PENDING · being built (no evidence yet)
code-server	SSH-key git access from editor terminal (host ~/.ssh)	e2e, challenges, helixqa	PENDING	Impl: In progress · Video: PENDING · being built (no evidence yet)
code-server	.bashrc / profile sourced in login-shell terminal	e2e, challenges	PENDING	Impl: In progress · Video: PENDING · being built (no evidence yet)
code-server	Editor Explorer defaults to \$PROJECTS_ROOT (convenience view, NOT a jail — terminal/ Open Folder/ extensions retain full host access by design)	e2e, challenges, helixqa	PENDING	Impl: In progress · Video: PENDING · being built (no evidence yet)
helix-auth	Auth fails CLOSED (gate down/errored → access denied, never bypassed)	stress_chaos, security	PENDING	Impl: In progress · Video: PENDING · being built (no evidence yet)

Honest boundaries (§11.4.6)

- Real-account + SSH-key authentication (the five rows above) is **in progress and unvalidated** — the helix-auth gate, the challenge/verify flow, and the auth / fail-closed / cookie / rate-limit tests are still being built. No PASS is claimed until captured evidence lands. (This supersedes the earlier PAM login: non-root PAM verify is impossible on this ALT/tcb host — see the pivot spec.)
- The editor Explorer defaulting to \$PROJECTS_ROOT is a **convenience view, NOT a security boundary**: the integrated terminal, File > Open Folder, and extensions retain full real-user (milosvasic) host-filesystem access by design (no code-server flag confines the process). This is the operator's chosen posture; real isolation would need a container / VM / chroot (out of scope). Defense-in-depth (deploy stream, in progress):
 - disable-workspace-trust
 - /proxy/ blocked at the Caddy edge.
- Real, publicly-trusted Let's Encrypt cannot be autonomously proven on a LAN box (no public domain); the ACME issue/serve/rotate code path is proven against a local Pebble CA, and the operator enables the public CA per docs/guides/TLS.md.
- Full in-browser editor interaction (typing, file tree, command palette) is operator_attended until a browser-automation adapter is wired (HC-QA-UI-001); every HTTP-drivable journey is autonomously verified.